MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5665

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, inert gas

See figure 1

Mounting position: Vertical, base down

Weight: 1-pound (453.6 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	epy	Ib	ib	i surge 0.1 sec	tk	TA
Unit:	V (see note 1)	v	v	Adc	a	aac	sec	°C
Maximum:	2.625	1,250	1,000	16	160	1,000	60	+70
Minimum:	2.375	---	---	---	---	---	---	-55
TEST CONDITIONS:	2.5	1,250	1,000	---	---	---	---	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3241	Heater current	-		0.65	II	If	28	34	Aac
3201	Critical grid voltage for conduction	-	Ebb = 750 Vdc; Rg = 10,000 ohms (max); Rp = 750 to 5,000 ohms	0.65	II	Ecc	-1.8	-6.2	Vdc
3201	Critical anode voltage for conduction	-	Ecc = +3 Vdc; Rg = 10,000 ohms (max); Rp = 750 to 5,000 ohms	0.65	II	Ebb	---	75	Vdc
3204	Voltage drop	2	Ib = 16 Adc	0.65	II	Etd	---	14	Vdc
1231	Peak emission by voltage drop	-	ib = 160 a; grid connected to anode thru 10,000 ohms	0.65	II	etd	---	60	v
	Quality conformance inspection, part 2								
1042	Shock, specified pulse	4	Test condition J; no voltages	---	---	---	---	---	---
3206	Operation	-	Ib = 16 Adc; Rg = 10,000 ohms	---	---	Ecc	-1.8	-6.2	Vdc
3216	Grid current	3	Rg = 1 Meg (min) and 10,000 ohms	---	---	Ic	---	10	uAdc
	Quality conformance inspection, part 3								
---	Life test	-	Group D, Epp = 220 Vac; Ib = 16 Adc; ib = 160 a; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission by voltage drop	-		---	---	etd	---	74	v
3216	Grid current	-		---	---	Ic	---	10	uAdc
3201	Critical grid voltage for conduction	-		---	---	Ecc	-1.8	-6.2	Vdc

NOTES:

1. Filament lead marked "F-" shall be phased negative when anode is positive.
2. This test is to be the first test performed at the conclusion of the holding period.
3. Operate tube at least 5 minutes prior to test. Operation may be at optional Epp but Ib must be 16 Adc minimum. To test, switch to test voltage Epp = 710 Vac and Ib = 16 Adc minimum in less than 2 seconds.
4. At the conclusion of this test, the TUT shall satisfy the requirements of method 1231 and method 3206 in quality conformance inspection, part 1 and part 2 respectively.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

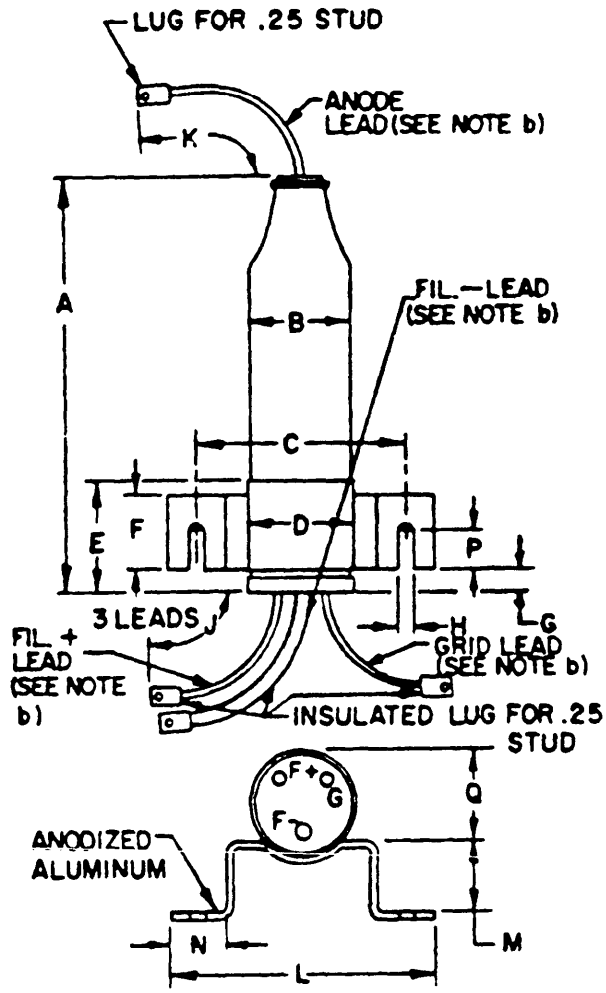
User activities:
Army - AR
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3188)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	9.500 (241.30)	10.500 (266.70)
B	2.375 (60.32)	2.750 (69.85)
C	4.875 (123.82)	5.125 (130.18)
M	1.437 (36.50)	1.687 (42.85)
Quality conformance inspection, part 3 (see note d)		
D	2.562 (65.07)	2.812 (71.42)
E	2.562 (65.07)	2.687 (68.25)
F	1.687 (42.85)	1.812 (46.02)
G	.437 (11.10)	.562 (14.27)
H	.312 (7.92)	.437 (11.10)
J	5.375 (136.52)	5.875 (149.22)
K	7.250 (184.15)	7.750 (196.85)
L	6.375 (161.92)	6.625 (168.28)
N	1.562 (39.67)	1.812 (46.02)
P	.812 (20.62)	.937 (23.80)
Q	2.187 (55.55)	2.437 (61.90)

NOTES:
a. Use of ceramic beads or fiber glass sleeves with silicone resin impregnation as lead insulators is optional.
b. Leads: Flexible; silicone-impregnated glass fiber insulation, class C-2 or better; as follows: (see notes c and e)
   Anode: at top; lug for .25 inch nominal (6.35 mm) stud, lug width .41 inch nominal (10.4 mm).
   Fil. + : at base; insulated lug for .25 inch nominal (6.4 mm) stud, lug width .47 inch nominal (11.9 mm).
   Fil. - : at base; insulated lug for .25 inch nominal (6.4 mm) stud, lug width .41 inch nominal (10.4 mm).
   Grid: at base; insulated lug for .25 inch nominal (6.4 mm) stud, lug width .47 inch nominal (11.9 mm).
c. Each lead shall be clearly marked as specified herein.
d. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
e. Nominal dimensions are for reference purposes only, and are not for inspection.

Ⓓ FIGURE 1. Outline drawing of electron tube type 5665.

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